



Cyclistic Bike-Share Analysis

Understanding Rider Behavior to Support Membership Conversion

Data Analyst Portfolio Project
Case Study & Strategic Recommendations



The Conversion Imperative



The Business Challenge

Cyclistic's future growth depends on maximizing annual memberships.

Currently, a high volume of the user base relies on single-ride or day-pass options, representing untapped recurring revenue.



The Analytical Objective

Analyze distinct usage patterns between annual members and casual riders.

Uncover actionable behavioral insights to design data-driven marketing strategies that convert casual riders into profitable annual members.

Data Infrastructure & Tooling

Historical Cyclistic Trip Data

783,799

valid trip records

Annual Members

Casual Riders



Python
(Core analysis)



Pandas



NumPy



Matplotlib



Seaborn

The Analytics Methodology



Ask

Define conversion objectives.



Prepare

Gather historical trip data.

The data used in this project is data for the period Q4 of 2022



Process

Clean 783K+ rows for integrity.



Analyze

Identify behavioral divergence.



Share

Visualize findings for alignment.



Act

Deliver strategic recommendations.

This presentation focuses on synthesizing analysis into shared action.

Executive Summary: A Tale of Two Riders

Volume

91.4%

of total rides (716,406) are generated by Annual Members, vs. 67,393 by Casual Riders.

Duration

3.4x

Casual rider trips are 3.4x longer on average (38.49 min vs 11.47 min).

Weekday Pattern

83.43%

of all Member rides occur on weekdays.

Weekend Pattern

47.35%

of Casual rides occur on weekends (compared to only 16.57% for members).

Rider Archetypes: Routine vs. Recreation



Annual Members

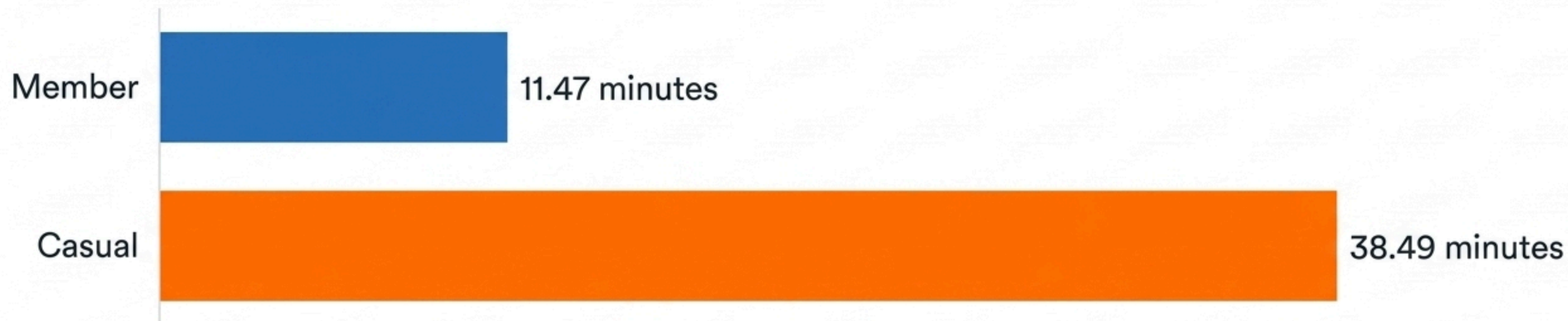
- ✓ **Usage Intent:** Commuting & Routine transportation
- ✓ **Frequency:** High volume, recurring
- ✓ **Duration:** Short, efficient trips
- ✓ **Peak Times:** Weekdays (8 AM & 5 PM)



Casual Riders

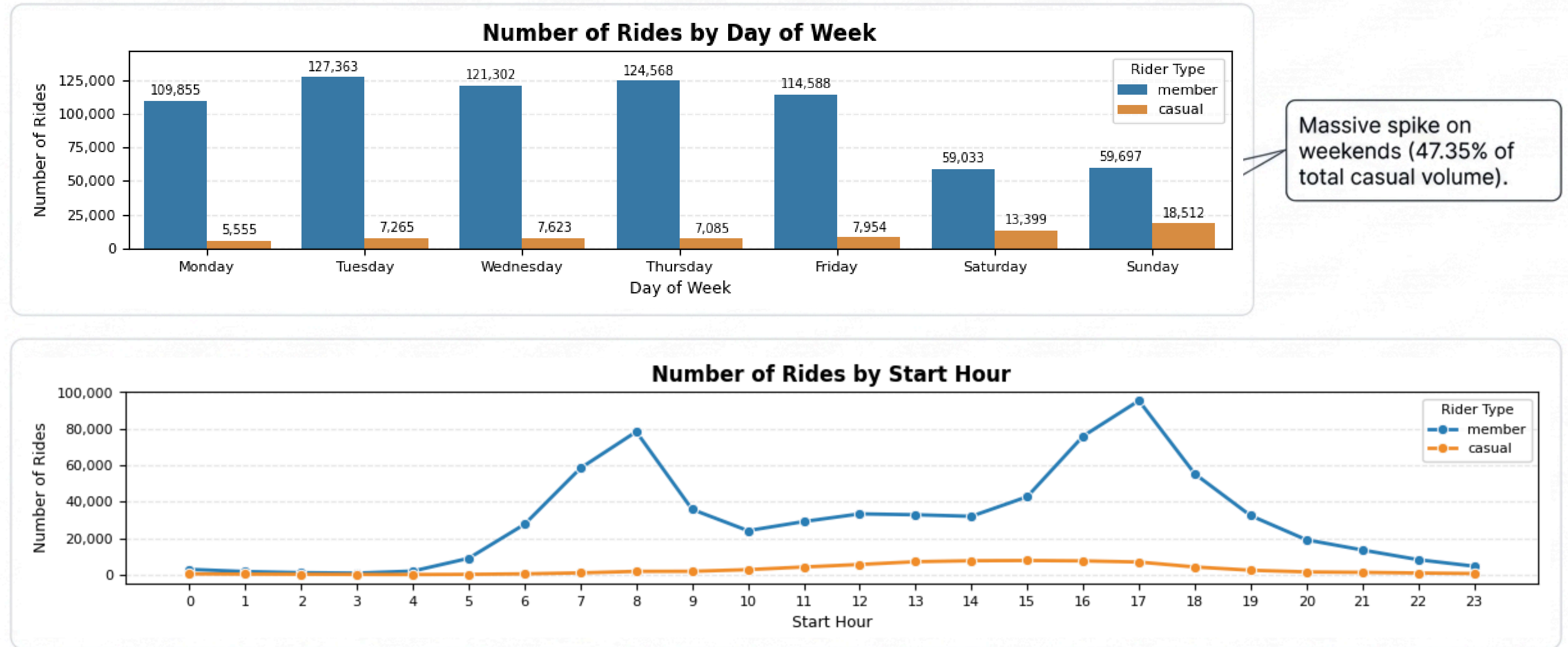
- ✓ **Usage Intent:** Leisure, tourism, & recreation
- ✓ **Frequency:** Lower volume, sporadic
- ✓ **Duration:** Long, extended rides
- ✓ **Peak Times:** Weekends & Afternoons

Casual Riders Optimize for Time, Not Frequency



A casual trip lasts approximately 27 minutes longer than a member trip on average. While members use bikes as a quick point-A-to-point-B utility, casual riders are heavily invested in the experience of the ride itself.

Temporal Divergence: When Riders Connect



Members are dictated by the traditional 9-to-5 commute, while Casual riders organically peak during leisure hours.

From Data to Opportunity

Observation

The data reveals a rigid divide—Members dominate weekday utility (short, frequent), while Casuals dominate weekend leisure (long, sporadic).



Interpretation

Casual riders are not “failed members”; they perceive value differently. They optimize for experience and time rather than daily commuter utility.



Opportunity

To convert Casuals, Cyclistic must aggressively reframe the Annual Membership. It must be positioned not just as a commuter tool, but as a hybrid pass unlocking both seamless commutes and limitless weekend recreation.

Strategic Conversion Roadmap



1. Targeted Personalization

Identify frequent casual riders within the database; target them with personalized, data-backed membership offers based on their specific ride history.



2. Timing Optimization

Concentrate digital marketing and promotional campaign spend strictly on **weekends** and **late afternoon peak hours** to intercept casuals when active.



3. Value Repositioning

Launch creative campaigns highlighting membership as the **ultimate dual-purpose tool** (Commuting + Recreation).



4. Frictionless Entry

Test introductory weekend-specific discount models to bridge the conversion gap; rigorously measure subsequent conversion and retention rates.

ROI of Data-Driven Strategy

**Weekend-Targeted
Ad Spend**



Maximizes marketing ROI by reaching Casual riders exactly when their intent and activity are highest.

**Repositioned Value
Proposition**



Increases Casual-to-Member conversion rates by aligning the product with their leisure-first mindset.

**Introductory
Discount Testing**



Improves customer acquisition while building long-term, predictable recurring revenue through retention.

“Data-driven insights
transform rider behavior into
growth opportunities.”